

Paper 1 Section A

Question No.	Key	Question No.	Key
1.	C (88)	26.	B (51)
2.	D (55)	27.	A (33)
3.	B (33)	28.	A (56)
4.	C (50)	29.	B (53)
5.	C (75)	30.	A (48)
6.	C (52)	31.	B (59)
7.	C (63)	32.	B (65)
8.	A (58)	33.	D (72)
9.	D (48)		
10.	A (58)		
11.	B (63)		
12.	A (77)		
13.	B (25)		
14.	D (50)		
15.	B (58)		
16.	B (66)		
17.	A (64)		
18.	D (37)		
19.	C (67)		
20.	D (46)		
21.	C (62)		
22.	D (78)		
23.	D (63)		
24.	A (66)		
25.	C (57)		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

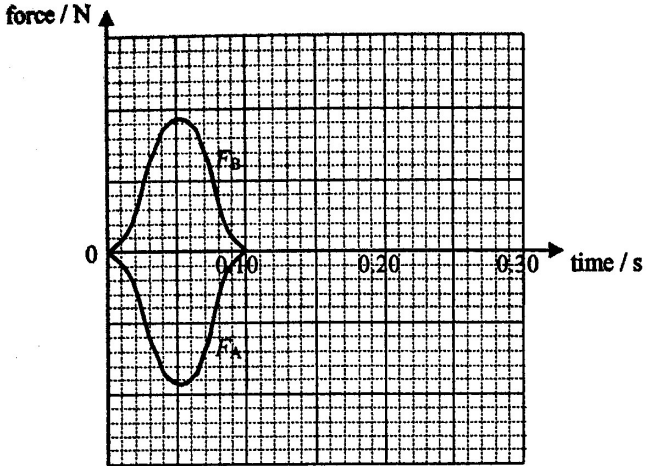
Paper 1 Section B

Solution	Marks	Remarks
1. (a) The glass cup absorbs thermal energy (from the hot water) / Heat is lost to the glass cup.	1A	
	1	
(b) $4200 \times 0.4 \times (100 - 93) = H \times (93 - 23)$ $H = 168 \text{ (J } ^\circ\text{C}^{-1}) \approx 170 \text{ (J } ^\circ\text{C}^{-1})$	1M 1A	
	2	
(c) (i) Infra-red (radiation) / thermal radiation	1A	
	1	
(ii) Remote control / infra-red (IR) thermometer / night vision camera	1A	
	1	
2. (a) $s = ut + \frac{1}{2}at^2$ $0 = 5t + \frac{1}{2}(-9.81)t^2$ $t = 1.0194 \text{ s} \approx 1.02 \text{ s}$ or 0 s (rejected)	1M 1A	
	2	
(b) speed of the bead at its maximum height = 2 m s^{-1} $\text{K.E.} = \frac{1}{2} \times 0.01 \times 2^2$ $= 0.02 \text{ J}$	1M 1A	
	2	
(c) The bead and the trolley have the same horizontal velocity and thus travel the same horizontal distance (within the same time interval t).	1A 1A	
	2	

Or
 $v = u + at,$
 $0 = 5 + (-9.81)(t/2)$
 $t = 1.02 \text{ s}$

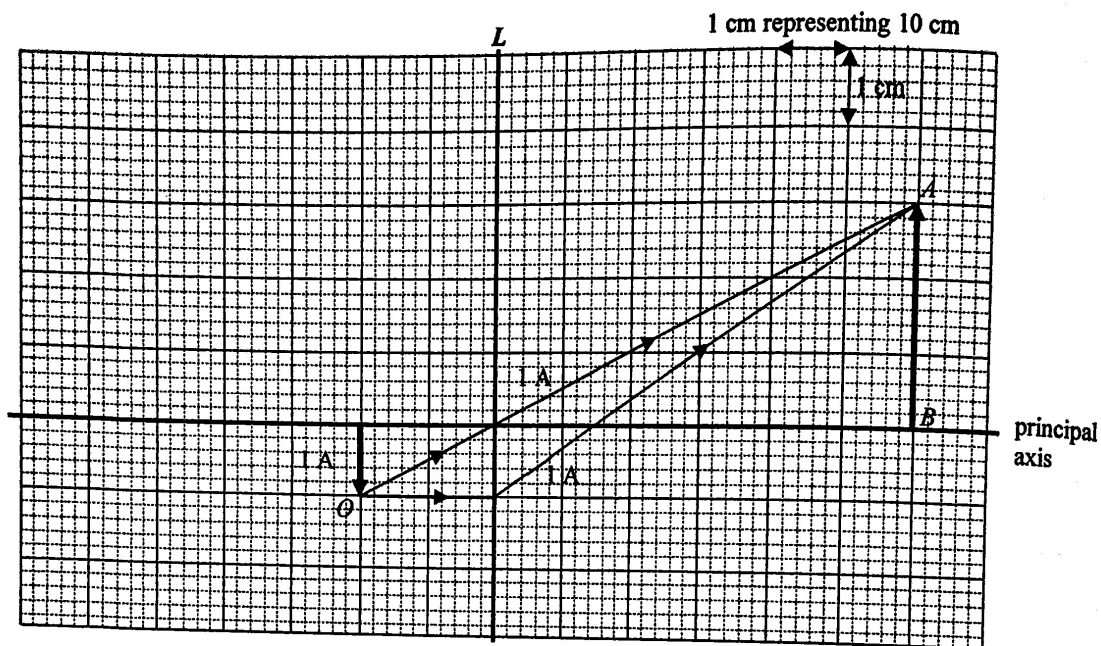
Accept: $t = 1.00 \sim 1.02 \text{ s}$

Solution	Marks	Remarks
3. (a) The object is moving to the left with constant deceleration and (momentarily) stops at $t = 4$ s. (b) Displacement = total area of the graph $= \frac{1}{2} \times (-10) \times 4 + \frac{1}{2} \times 15 \times (10 - 4)$ $= 25$ m Position of the object is 25 m from point O on the right side. (c) Net force acting on the object $= ma$ $= 0.4 \times \frac{15 - (-10)}{10 - 0}$ $= 1$ N (towards right)	1A 1A 2 1M 1A 2 1M 1A 2	Or $v^2 = u^2 + 2as$ $15^2 = (-10)^2 + 2 \times \frac{15 - (-10)}{10 - 0} \times s$ $s = 25$ m
4. (a) $\omega = \frac{2\pi}{60}$ $= \frac{\pi}{30}$ or 0.105 (rad s⁻¹) (b) (c) The required centripetal force $F_C = m\omega^2 r$, where r is the distance of plasticine Q from the clock's centre. As m and ω are constant, smaller r gives smaller centripetal force F_C. The claim is correct.	1A 1 1A 1 1A+1A 2	Accept: adhesive force

Solution	Marks	Remarks
5. With the computer interface switched on for data recording (force towards the right taken as positive), give trolley <i>A</i> a push and let it collide head-on with trolley <i>B</i>. Force-time graphs:  The force acting on (the less massive) trolley <i>B</i>, i.e. F_B, is always equal (in magnitude and opposite in direction) to that acting on trolley <i>A</i>, i.e. F_A, as they are an action and reaction pair / according to Newton's 3rd law.	1A 2A 1A 4	

Solution	Marks	Remarks
6. (a) Convex lens because only a convex lens can form a real image.	1A 1A	Or the image can be captured on a screen / the image is formed on the opposite side of the lens / only convex lens can form a magnified image.
	2	

(b) (i)(ii)



(iii) Ray diagram method:
 $f = 15 \text{ cm}$

3
1A
1
1A
1
1A
1

Accept: 14 cm ~ 16 cm

Or

By $\frac{v}{u} = 3$,

$$\frac{0.6}{u} = 3, \text{ i.e. } u = 0.2 \text{ m}$$

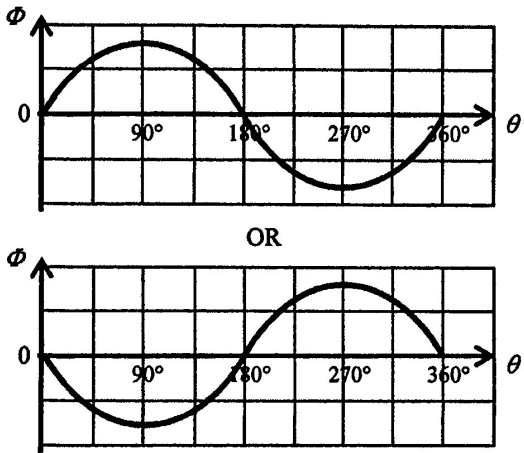
By $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$,

$$\frac{1}{f} = \frac{1}{0.2} + \frac{1}{0.6}$$

$$f = 0.15 \text{ m or } 15 \text{ cm}$$

(c) Move / adjust the support / smartphone closer to lens L.

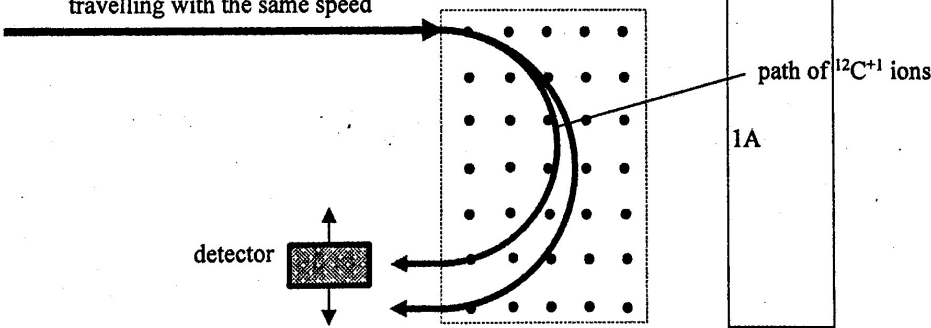
(d) The box should be painted black inside.

Solution	Marks	Remarks
7. (a) To protect the ammeter / limiting the current in the circuit / ensuring the circuit does not over-heat. (b) $R = \frac{V}{I} = \frac{3.5}{0.7}$ $= 5.0 \Omega (> 4.7 \Omega)$ The extra resistance (0.3Ω) may be due to the resistance of the ammeter / contact resistances / internal resistance of the power supply. (c) $I = 0.25 \text{ A}$ at $l = 30 \text{ cm}$ $\Delta R = R_{AC} = \frac{3.5}{0.25} - 5.0$ $= 9.0 \Omega$	1A 1 1M 1A 1A 3 1M 1A 2	Accept: $I = 0.24 \text{ A} \sim 0.26 \text{ A}$ e.c.f. from (b) Accept: $\Delta R = 8.5 \Omega \sim 9.6 \Omega$
8. (a)  OR (b) (i) $\phi = BA \cos 0^\circ = 2.70 \times 10^{-4}$ $B \times (9.00 \times 10^{-4}) \times 1 = 2.70 \times 10^{-4}$ $B = 0.300 \text{ (T)}$ (ii) $\mathcal{E} = \left -N \frac{\Delta \phi}{\Delta t} \right$ $= \left -100 \times \frac{2.70 \times 10^{-4}}{0.01} \right$ $= 2.70 \text{ V}$	2A 2 1M 1A 2 1M 1A 2	Accept: -2.70 V

Solution	Marks	Remarks
9. (a) (i) $pV = nRT$ $1.01 \times 10^5 \times V = 1 \times 8.31 \times 273$ $V = 2.2462 \times 10^{-2} \text{ m}^3 \approx 2.25 \times 10^{-2} \text{ m}^3$	1M 1A 2	Accept: 0.0220 ~ 0.0225 m ³
(ii) Number of molecules in 1 m ³ $= \frac{N_A}{\text{volume of one mole of molecule}}$ $= \frac{6.02 \times 10^{23}}{2.246 \times 10^{-2}}$ $= 2.680120 \times 10^{25} \approx 2.68 \times 10^{25}$	1M 1A 2	e.c.f. from (a)(i) Or $pV = nRT$ $1.01 \times 10^5 \times 1 = n \times 8.31 \times 273$ $n = 44.5$ $N = 44.5 \times 6.02 \times 10^{23}$ $= 2.68 \times 10^{25}$
(b) $pV = \left(\frac{N}{N_A} \right) RT$ $2.75 \times 10^{-10} \times 1 = \left(\frac{1.00 \times 10^{11}}{6.02 \times 10^{23}} \right) \times 8.31 \times T$ $T = 199.2178 \text{ (K)} \approx 199 \text{ (K)}$	1M 1A 2	Accept: 199 ~ 200 K

Solution	Marks	Remarks
10. (a) (i) 90 cm (ii) Particle X vibrates with a larger amplitude than particle Y while they are vibrating in phase. (iii) Correct sketch of a straight string (iv) Original wavelength $\lambda_1 = \frac{0.90}{1.5} = 0.60 \text{ m at } f_1 = 100 \text{ Hz}$ For the next stationary wave pattern, $\lambda_2 = \frac{0.90}{2} = 0.45 \text{ m}$ By $v = f\lambda$, $100 \times 0.60 = f_2 \times 0.45$ $f_2 = 133.33 \text{ Hz} \approx 130 \text{ Hz}$ (b) (i) (ii) The relative strengths and proportions of the harmonics determine the unique sound characteristics of different instruments, when these harmonics superimpose and combine to produce different waveforms. (iii) Any TWO: stationary vs travelling transverse vs longitudinal energy confined in a string vs energy being transferred through air	2A 2 1A 1A 2 1A 1 1M 1A 2 1A 1 1A 1A 2 2A 2	Or $f_2 = 100 \times \frac{4}{3}$ $= 133.33 \text{ Hz} \approx 130 \text{ Hz}$

Solution		Marks	Remarks
11. (a)	(i) $E_{in} = (32 \times 10^6) \times (4.5 \times 10^6)$ $= 1.44 \times 10^{14} \text{ J} \approx 1.4 \times 10^8 \text{ (MJ)}$	1M 1A 2	e.c.f. from (a)(i)(ii) Accept: 41% ~ 42%
	(ii) $E_{out} = (680 \times 10^6) \times (24 \times 3600)$ $= 5.875 \times 10^{13} \text{ J} \approx 5.9 \times 10^7 \text{ (MJ)}$	1M 1A 2	
	(iii) Efficiency = $\frac{E_{out}}{E_{in}} \times 100\%$ $= \frac{5.88 \times 10^{13}}{1.44 \times 10^{14}} \times 100\%$ $= 40.8\% \approx 41\%$	1M 1A 2	
	(b) (i) $P = VI$ $680 \text{ MW} = 400 \text{ kV} \times I$ $I = 1700 \text{ A} = 1.7 \text{ kA}$	1M 1A 2	
	(ii) Power loss = $I^2 R$ $= (1.7 \times 10^3)^2 \times (2.90 \times 10^{-8} \times 160 \times 10^3)$ $= 13410 \text{ W} \approx 13.4 \text{ (kW)}$	1M 1A 2	
	(iii) Only a.c. voltage can be stepped up (or stepped down) using a transformer, which is cost effective. High voltage transmission has low transmission current and so reduces power loss significantly.	1A 1A 2	

Solution	Marks	Remarks
12. (a) (i) ${}^{14}_6\text{C} \rightarrow {}^{14}_7\text{N} + {}^0_{-1}\text{e}$ (or β)	1A	
	1	
(ii) $k = \frac{\ln 2}{t_{\frac{1}{2}}}$ $= \frac{\ln 2}{5370}$ $= 1.2097 \times 10^{-4} \text{ (yr}^{-1}\text{)} \approx 1.21 \times 10^{-4} \text{ (yr}^{-1}\text{)}$	1M 1A	
	2	
(iii) $C = C_0 e^{-kt}$ $6.0 = 13.6 e^{-(1.21 \times 10^{-4})t}$ $t = 6764.7 \approx 6800 \text{ (years)}$	1M 1A	e.c.f. from (a)(ii) Accept: 6760 ~ 6820 (years)
	2	
(b) (i) An isotope is a variation of an element that possesses the same atomic number but a different mass number.	1A	Or same number of protons but different number of neutrons
	1	
(ii) ${}^{12}\text{C}^{+1}$ and ${}^{14}\text{C}^{+1}$ ions travelling with the same speed		
	1A	
	1	
(iii) As the magnetic force induced on a carbon ion is always perpendicular to its velocity / direction of motion, no work is done on the ions.	1A 1A	
	2	